A ten-variable Diophantine candidate

Strictly positive integer unknowns; exact degree 750; 80 evaluator operations.

The only unknowns are $a, b, c, d, e, f, g, h, i, j \in \mathbb{Z}_{>0}$. Each $\texttt{r01}, \ldots, \texttt{r80}$ below is an abbreviation, not an additional unknown. Evaluate the rows in order using ordinary integer arithmetic; subtraction is integer subtraction, and intermediate values may be negative. The equation is $\boxed{\texttt{r80} = 0}$.

$\texttt{r01} = \beta \cdot a$
$\texttt{r02} = \texttt{r01} + 1$
$\texttt{r03} = \texttt{r02} \cdot \texttt{r02}$
$\texttt{r04} = \texttt{r03} \cdot \texttt{r02}$
$\texttt{r05} = \texttt{r04} \cdot g$
$\texttt{r06} = \texttt{r02} - b$
$\texttt{r07} = \texttt{r06} - c$
$\texttt{r08} = f \cdot \texttt{r02}$
$\texttt{r09} = i + j$
$\texttt{r10} = \texttt{r09} \cdot \texttt{r05}$
$\texttt{r11} = \texttt{r10} + i$
$\texttt{r12} = \texttt{r03} + 1$
$\texttt{r13} = \texttt{r08} + 1$
$\texttt{r14} = \texttt{r12} \cdot \texttt{r05}$
$\texttt{r15} = \texttt{r14} \cdot \texttt{r13}$
$\texttt{r16} = 4 \cdot \texttt{r15}$
$\texttt{r17} = \texttt{r16} - 1$
$\texttt{r18} = \texttt{r15} + 2$
$\texttt{r19} = \texttt{r14} \cdot \texttt{r14}$
$\texttt{r20} = \texttt{r08} \cdot \texttt{r19}$
$\texttt{r21} = \texttt{r20} \cdot \texttt{r09}$
$\texttt{r22} = \texttt{r21} + \texttt{r09}$
$\texttt{r23} = \texttt{r21} \cdot \texttt{r22}$
$\texttt{r24} = \texttt{r23} + 1$
$\texttt{r25} = \texttt{r01} \cdot \texttt{r06}$
$\texttt{r26} = \texttt{r25} + c$
$\texttt{r27} = \texttt{r03} - 1$
$\texttt{r28} = \texttt{r26} \cdot \texttt{r27}$
$\texttt{r29} = \texttt{r02} \cdot \nu$
$\texttt{r30} = \mu + \texttt{r29}$
$\texttt{r31} = \texttt{r30} \cdot a$
$\texttt{r32} = \texttt{r28} + \texttt{r31}$
$\texttt{r33} = \texttt{r09} - \texttt{r32}$
$\texttt{r34} = \texttt{r33} - 1$
$\texttt{r35} = \kappa + f$
$\texttt{r36} = \texttt{r35} \cdot \texttt{r07}$
$\texttt{r37} = \texttt{r36} - b$
$\texttt{r38} = \texttt{r11} \cdot \texttt{r11}$
$\texttt{r39} = d \cdot \texttt{r38}$
$\texttt{r40} = \texttt{r39} \cdot \texttt{r24}$
$\texttt{r41} = \texttt{r15} \cdot \texttt{r18}$
$\texttt{r42} = \texttt{r41} \cdot \texttt{r40}$
$\texttt{r43} = \texttt{r42} \cdot \texttt{r39}$
$\texttt{r44} = \texttt{r43} + 1$
$\texttt{r45} = \texttt{r43} \cdot \texttt{r44}$
$\texttt{r46} = 4 \cdot \texttt{r45}$
$\texttt{r47} = \texttt{r46} + 1$
$\texttt{r48} = \texttt{r41} \cdot \texttt{r46}$
$\texttt{r49} = \texttt{r11} \cdot e$
$\texttt{r50} = \texttt{r49} - \texttt{r32}$
$\texttt{r51} = \texttt{r47} \cdot \texttt{r50}$
$\texttt{r52} = \texttt{r51} - \texttt{r11}$
$\texttt{r53} = \texttt{r52} \cdot \texttt{r52}$
$\texttt{r54} = \texttt{r12} \cdot \texttt{r12}$
$\texttt{r55} = \texttt{r01} \cdot \texttt{r54}$
$\texttt{r56} = \texttt{r53} \cdot \texttt{r37}$
$\texttt{r57} = h - \texttt{r56}$
$\texttt{r58} = \texttt{r17} \cdot \texttt{r57}$
$\texttt{r59} = 3 \cdot \texttt{r11}$
$\texttt{r60} = 2 \cdot \texttt{r08}$
$\texttt{r61} = \texttt{r59} - \texttt{r60}$
$\texttt{r62} = \texttt{r08} \cdot \texttt{r61}$
$\texttt{r63} = \texttt{r62} + 2$
$\texttt{r64} = \texttt{r58} + \texttt{r63}$
$\texttt{r65} = \texttt{r55} \cdot \texttt{r64}$
$\texttt{r66} = \texttt{r54} \cdot \texttt{r37}$
$\texttt{r67} = \texttt{r01} \cdot \texttt{r34}$
$\texttt{r68} = \texttt{r66} + \texttt{r67}$
$\texttt{r69} = \texttt{r17} \cdot \texttt{r68}$
$\texttt{r70} = \texttt{r65} - \texttt{r69}$
$\texttt{r71} = \texttt{r55} \cdot \texttt{r17}$
$\texttt{r72} = \texttt{r71} \cdot \texttt{r71}$
$\texttt{r73} = \texttt{r24} \cdot \texttt{r44}$
$\texttt{r74} = \texttt{r70} \cdot \texttt{r70}$
$\texttt{r75} = \texttt{r72} \cdot \texttt{r73}$
$\texttt{r76} = \texttt{r75} \cdot \texttt{r53}$
$\texttt{r77} = \texttt{r74} - \texttt{r76}$
$\texttt{r78} = \texttt{r48} \cdot \texttt{r77}$
$\texttt{r79} = \texttt{r75} - \texttt{r74}$
$\texttt{r80} = \texttt{r78} + \texttt{r79}$

The four fixed integers. All indices in the following formulas are locally bound indices for constants, not unknowns. Set $\rho = 2^{67}$ and $\beta = \rho^{8807} - 1$. Let the fixed set and ordered list be

$$\begin{aligned}\mathcal{S} = \big(&\{0,\ldots,942\} \cup \{983,\ldots,992\} \cup \{1003,\ldots,1011\} \cup \{1013,\ldots,1031\} \\ &\cup \{1063, 1064, 1095, 2195, 3296\}\big) \setminus \{1\},\\ \lambda = (&1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014,\\ &1015, 1016, 1017, 1018, 1019, 1020, 1021, 1003, 1004, 1005,\\ &1006, 1007, 1008, 1009, 1010, 983, 984, 985, 986, 987,\\ &988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195,\\ &3296).\end{aligned}$$

The list λ has 41 entries, numbered from zero. The fixed list σ has 943 entries, also numbered from zero, and is printed in decimal below and on the following pages. Then

$$\begin{aligned}\mu &= \beta - \sum_{u\in\mathcal{S}} \rho^u - 2\rho^{1063},\\ \nu &= \beta + 4 + \sum_{u=1013}^{1022} \rho^u + \sum_{u=1065}^{1073} \rho^u + \rho^{2195} + \rho^{3296}\\ &\quad + (2^{51}-2)(\rho^{4393} + \rho^{5495}) + (2^{65}-1)\sum_{u=8792}^{8804} \rho^u + (2^{19}-1)\rho^{8805},\\ \kappa &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849}\\ &\quad + \sum_{u=0}^{942} \sigma_u(\rho^{4393-u} + \rho^{5495-u}) + \sum_{u=0}^{40} 2^{10+u}(\rho^{4393-\lambda_u} + \rho^{5495-\lambda_u}).\end{aligned}$$

Decimal selector list. Read from left to right, top to bottom; each 14-digit block is one decimal integer. Leading zeroes do not change its value. There are exactly 943 blocks.

```
17557309618177 17592186039297 17592186044417 02542620639233 17427232456705 17358378762241 17289525067777
17220671373313 17151817678849 17014244507649 16876671336449 16807817641985 16738963947521 16463951822849
16395098128385 16120086003713 15913793355777 15707500707841 15569927536641 15432354365441 15363500670977
15157208023041 15088354328577 14744622727169 14607049555969 14057159524353 13782147399681 12888659984385
12476208906241 12338635735041 11994904133633 11307575148545 10895124070401 10001636655105 09932782960641
09863929266177 09314039234561 08832868679681 08351698124801 08282844430337 07664234921985 07389222797313
07114210672641 06907918024705 06151869562881 05945576914945 04296175255553 03608846270465 02784078331905
17557826043905 02542620377089 17392872456193 17324018761729 17255165067265 17186311372801 17117457678337
16979884507137 16842311335937 16773457641473 16704603947009 16429591822337 16360738127873 16085726003201
15879433355265 15673140707329 15535567536129 15397994364929 15329140670465 15122848022529 15053994328065
14710262726657 14572689555457 14022799523841 13747787399169 12854299983873 12441848905729 12304275734529
11960544133121 11273215148033 10860764069889 09967276654593 09898422960129 09829569265665 09279679234049
08798508679169 08317338124289 08248484429825 07629874921473 07354862796801 07079850672129 06873558024193
06117509562369 05911216914433 04261815255041 03574486269953 02749718331393 17546148839425 17557020213249
17556884160513 17591242850305 17590839934977 17581578649601 17532991569921 17590302277633 17556076494849
17590837837825 17543726891009 17589763309569 17555269091329 17589359345665 17589223817217 17588149813249
17565735190529 17531509407745 17588685897729 17554191679489 17588416937985 17553922719745 17589221720065
17584926490625 17588013236225 17554995412993 17591367368705 17587743227905 17554725404673 17587608485889
17553114267649 17587339526145 17587204784129 17552710565889 17586935824385 17552441606145 17586666864641
17552172646401 17589484912641 17552037904385 17586263162881 17553782210561 17588141686785 17553647468545
17553512464385 17585992630273 17553377460225 17590018637825 17543444824065 17590420766721 17551363145729
17587735625729 17589211234305 17589076754433 17590418669569 17585452089345 17550957871105 17585183129601
17550688911361 17589209137153 17585584996353 17584779427841 17550285209601 17584510468097 17589073346561
17551894773761 17584374939649 17590145515521 17551624765441 17573367775233 17539544645633 17583836233729
17549342015489 17589338112001 17551220015105 17585981882369 17584773660673 17583565176833 17549070958593
17584638394369 17548936216577 17585577394177 17550546305025 17586113740801 17589066268673 17583025160193
17550007336961 17579937628161 17538732523521 17533766205441 17573762564097 17539268345857 17582620409857
17530409451521 17553897029633 17587182764033 17550138408961 17547990401025 17585436884993 17582080917505
17547586699265 17556981678081 17550002094081 17582885175297 17547451432961 17581407731713 17546913513473
17579796594689 17530672644097 17580869812225 17550938472449 17586103255041 17587982041089 17580599803905
17580330844161 17531877982209 17580061884417 17556170866689 17581000884225 17545567141889 17564894232577
17555096076289 17577375956993 17545163440129 17573080465409 17543552303105 17579253956609 17544759738369
17582072004609 17547846221825 17579387125761 17532142223361 17578849730561 17544355512321 17584083697665
```

17591331192833 17588780793857 17544220246017 17568379174913 17589718745089 17543816019969 17584752164865
17550392164353 17583677898753 17540593745921 17583945809921 17533077028865 17578845011969 17577233874945
17542739656705 17583273148417 17550657978369 17585151672321 17542604390401 17589446115329 17542469648385
17589445591041 17584613228545 17552669147137 17576693858305 17556560936961 17579377688577 17542199115777
17573068931073 17533474439169 17576289107969 17544613462017 17582731034625 17535217696769 17571053568001
17534277648385 17575750664193 17573200003073 17539108438017 17571723083777 17531457503233 17587291291649
17548770541569 17584606412801 17540985389057 17584605888513 17540850647041 17583397404673 17590241984513
17540983816193 17589436153857 17540714856449 17573329502209 17534674534401 17574671155201 17575475937281
17534405050369 17565274865665 17535746703361 17582722646017 17534940872705 17572253138945 17556146749441
17565407510529 17537758396417 17590505701377 17539771138049 17588357693441 17549165854721 17587283427329
17572653170689 17572518428673 17580973621249 17540439605249 17566343364609 17539902210049 17589159854081
17586206801921 17574261161985 17534398234625 17582447919105 17545806217217 17580031475713 17537484193793
17569427750913 17536544145409 17590499409921 17537483145217 17580969426945 17580968902657 17537616314369
17581504724993 17546607853569 17571840524289 17556136787969 17581637894145 17539493265409 17571168387073
17536271515649 17573180604417 17531170193409 17571837902849 17543249264641 17579890442241 17545530441729
17586063933441 17588076937217 17570762588161 17589955198977 17538012676097 17545662562305 17570626797569
17534521966593 17589685190657 17535863619585 17569686224897 17536668401665 17571833184257 17545660465153
17576127627265 17536801570817 17579885199361 17537069481985 17539619094529 17535323602945 17573441175553
17535457296385 17590620520449 17541228134401 17571561078785 17539482779649 17567534022657 17536663683073
17569009893377 17534649892865 17568606715905 17542165561345 17573572247553 17536662110209 17570484715521
17546727915521 17582698004481 17550351269889 17573570674689 17546592649217 17571556884481 17540015456257
17574240714753 17547128471553 17569408352257 17548335906817 17573702795265 17535584698369 17573970706433
17579875762177 17551421341697 17570614214657 17551420817409 17589941043201 17573834391553 17534776770561
17582558019585 17542963527681 17578665181185 17541889261569 17570074198017 17537459027969 17573026463745
17543095648257 17568462536705 17533968318465 17566985617409 17545242083329 17573427544065 17531685568513
17572084842497 17532758786049 17569399963649 17533563568129 17584163389441 17556648493057 17573022793729
17541481365505 17571143221249 17533293559809 17572753309697 17542017187841 17578926800897 17545640542209
17580671107073 17546579542017 17578388881409 17554497863681 17565100802049 17543894138881 17573019123713
17535974768641 17565905059841 17537316421633 17574226034689 17540134469633 17566575099905 17555434766337
17566306140161 17554226282497 17571271671809 17532482486273 17575968768001 17538521759745 17566707220481
17533823614977 17586570919937 17545097379841 02748665823233 02542507393025 02714305822721 02542507130881
02576865820673 02576731602945 02576597385217 02576865819649 02576731601921 02576597384193 02679810554881
02645450816513 02611091078145 17591803836417 11131904062465 17591534874625 02576865816577 02576731598849
02576597381121 17523466561537 17489106561025 17111149700097 17076789699585 17042430222337 17008070221825
16973710744577 16939350744065 16836271790081 16801911789569 16767552312321 16733192311809 16698832834561
16664472834049 16630113356801 16595753356289 16561393879041 16527033878529 16423954924545 16389594924033
16355235446785 16320875446273 16286515969025 16252155968513 16217796491265 16183436490753 16149077013505
16114717012993 16080357535745 16045997535233 16011638057985 15977278057473 15874199103489 15839839102977
15805479625729 15771119625217 15736760147969 15702400147457 15668040670209 15633680669697 15599321192449
15564961191937 15530601714689 15496241714177 15461882236929 15427522236417 15393162759169 15358802758657
15324443281409 15290083280897 15255723803649 15221363803137 15118284849153 15083924848641 15049565371393
15015205370881 14980845893633 14946485893121 14912126415873 14877766415361 14843406938113 14809046937601
14705967983617 14671607983105 14637248505857 14602888505345 14568529028097 14534169027585 14499809550337
14465449549825 14431090072577 14396730072065 14362370594817 14328010594305 14293651117057 14259291116545
14224931639297 14190571638785 14156212161537 14121852161025 14018773207041 13984413206529 13950053729281
13915693728769 13881334251521 13846974251009 13743895297025 13709535296513 13675175819265 13640815818753
13606456341505 13572096340993 13537736863745 13503376863233 13469017385985 13434657385473 13400297908225
13365937907713 13331578430465 13297218429953 13262858952705 13228498952193 13194139474945 13159779474433
13125419997185 13091059996673 13056700519425 13022340518913 12987981041665 12953621041153 12919261563905
12884901563393 12850542086145 12816182085633 12781822608385 12747462607873 12713103130625 12678743130113
12644383652865 12610023652353 12575664175105 12541304174593 12506944697345 12472584696833 12438225219585
12403865219073 12369505741825 12335145741313 12300786264065 12266426263553 12232066786305 12197706785793
12163347308545 12128987308033 12094627830785 12060267830273 12025908353025 11991548352513 11957188875265
11922828874753 11888469397505 11854109396993 11819749919745 11785389919233 11751030441985 11716670441473
11682310964225 11647950963713 11613591486465 11579231485953 11544872008705 11510512008193 11476152530945
11441792530433 11407433053185 11373073052673 11338713575425 11304353574913 11269994097665 11235634097153
11201274619905 11166914619393 11132555142145 11098195141633 11063835664385 11029475663873 10995116186625
10960756186113 10857677232129 10823317231617 10788957754369 10754597753857 10720238276609 10685878276097
10651518798849 10617158798337 10582799321089 10548439320577 10514079843329 10479719842817 10445360365569
10411000365057 10376640887809 10342280887297 10307921410049 10273561409537 10239201932289 10204841931777
10170482454529 10136122454017 10101762976769 10067402976257 10033043499009 09998683498497 09964324021249

09929964020737 09826885066753 09792525066241 09758165588993 09723805588481 09689446111233 09655086110721
09620726633473 09586366632961 09552007155713 09517647155201 09483287677953 09448927677441 09414568200193
09380208199681 09345848722433 09311488721921 09277129244673 09242769244161 09208409766913 09174049766401
09139690289153 09105330288641 09070970811393 09036610810881 09002251333633 08967891333121 08933531855873
08899171855361 08864812378113 08830452377601 08796092900353 08761732899841 08727373422593 08693013422081
08658653944833 08624293944321 08589934467073 08555574466561 08521214989313 08486854988801 08452495511553
08418135511041 08315056557057 08280696556545 08246337079297 08211977078785 08177617601537 08143257601025
08108898123777 08074538123265 08040178646017 08005818645505 07971459168257 07937099167745 07902739690497
07868379689985 07834020212737 07799660212225 07765300734977 07730940734465 07696581257217 07662221256705
07627861779457 07593501778945 07559142301697 07524782301185 07490422823937 07456062823425 07421703346177
07387343345665 07352983868417 07318623867905 07284264390657 07249904390145 07215544912897 07181184912385
07078105958401 07043745957889 07009386480641 06975026480129 06940667002881 06906307002369 06871947525121
06837587524609 06803228047361 06768868046849 06734508569601 06700148569089 06665789091841 06631429091329
06597069614081 06562709613569 06528350136321 06493990135809 06459630658561 06425270658049 06390911180801
06356551180289 06322191703041 06287831702529 06253472225281 06219112224769 06184752747521 06150392747009
06116033269761 06081673269249 06047313792001 06012953791489 05978594314241 05944234313729 05909874836481
05875514835969 05841155358721 05806795358209 05772435880961 05738075880449 05703716403201 05669356402689
05634996925441 05600636924929 05566277447681 05531917447169 05497557969921 05463197969409 05428838492161
05394478491649 05360119014401 05325759013889 05291399536641 05257039536129 05222680058881 05188320058369
05153960581121 05119600580609 05085241103361 05050881102849 05016521625601 04982161625089 04947802147841
04913442147329 04879082670081 04844722669569 04810363192321 04776003191809 04741643714561 04707283714049
04672924236801 04638564236289 04604204759041 04569844758529 04535485281281 04501125280769 04466765803521
04432405803009 04398046325761 04363686325249 04329326848001 04294966847489 04260607370241 04226247369729
04191887892481 04157527891969 04123168414721 04088808414209 04054448936961 04020088936449 03985729459201
03951369458689 03917009981441 03882649980929 03848290503681 03813930503169 03779571025921 03745211025409
03710851548161 03676491547649 03642132070401 03607772069889 03573412592641 03539052592129 03504693114881
03470333114369 03435973637121 03401613636609 03367254159361 03332894158849 03298534681601 03264174681089
03229815203841 03195455203329 03161095726081 03126735725569 03092376248321 03058016247809 03023656770561
02989296770049 02954937292801 02920577292289 02886217815041 02851857814529 02748778860545 02714418860033
17592184785921 17557817183233 17592176659457 17557806959617 17592123182081 17557824784385 17557824260097
17557823735809 17557822687233 17557820590081 17557817444353 17557813250049 17557807482881 17557799094273
17592184259585 17592182947841 17592164335617 17592073109505 17592183733249 17592098273281 17592182682625
17592129466369 17592177961985 17592164854785 17592155941889 17592180582401 17592180057089 17592163017729
17592160396289 17592155153409 17592121074689 17592114258945 17592113734657 17557759238145 17592177433601
17592179529729 17592152790017 17557792264193 17557818477569 17592173234177 17592172708865 17592159601665
17592170872833 17592156979201 17557726461953 17592142035969 17592167462913 17557804315649 17557796975617
17557790684161 17592105594881 17557792779265 17592159070209 17557787534337 17592178466817 17592148058113
17557774949377 17592126823425 17592143337473 17592097723393 17557775208449 17557726187521 17592152247297
17557755021313 17592123409409 17557728281601 17557783855105 17557785426945 17557745581057 17592145426433
17557729850369 17592108199937 17557748198401 17557749245953 17592101905409 17592149090305 17592135195649
17592132049921 17592127069185 17592124709889 17592102689793 17592072805377